INCH-POUND

MIL-DTL-83513/4G
25 June 2002
SUPERSEDING
MIL-PRF-83513/4F
15 August 1997

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, SOCKET CONTACTS, CLASS M, CRIMP TYPE,

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83513.

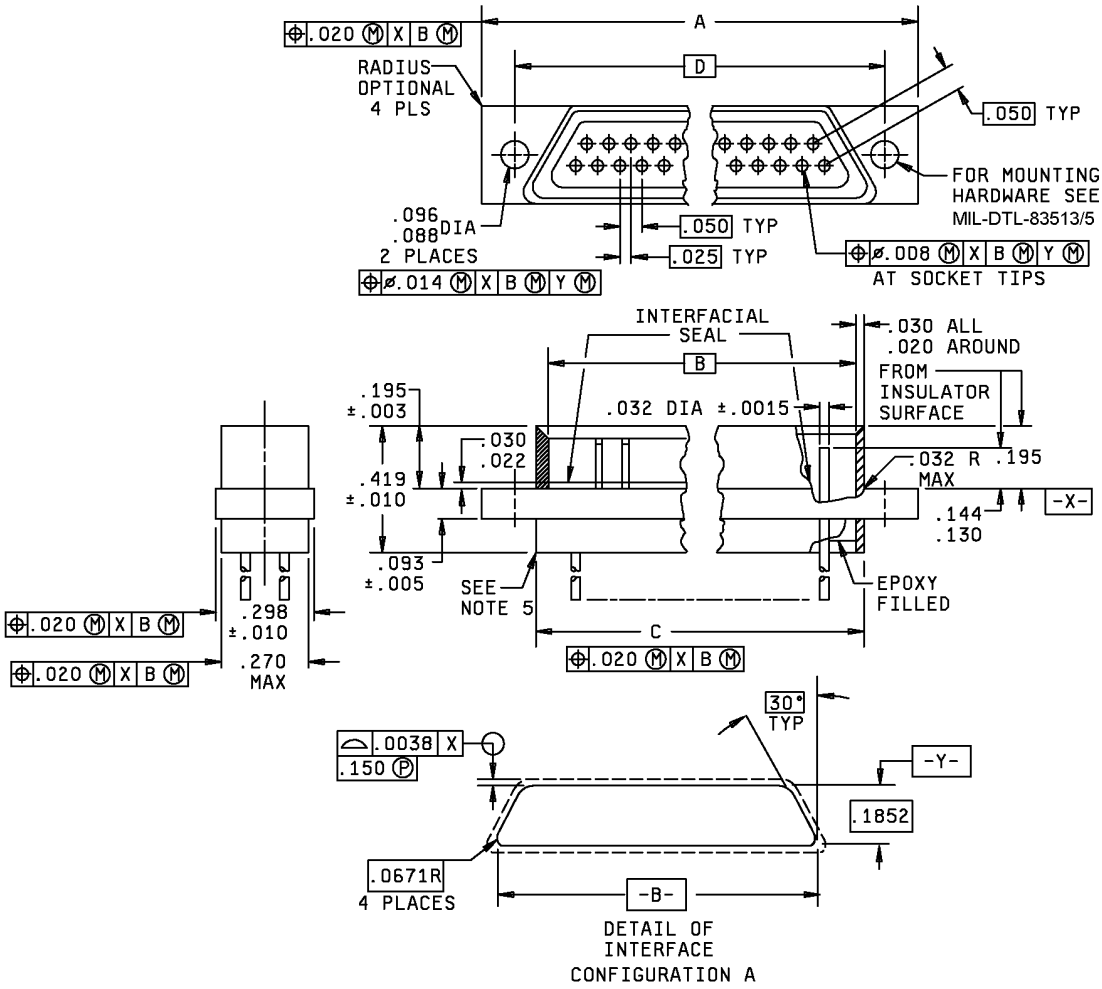


FIGURE 1. Connector, plug .050 spacing.

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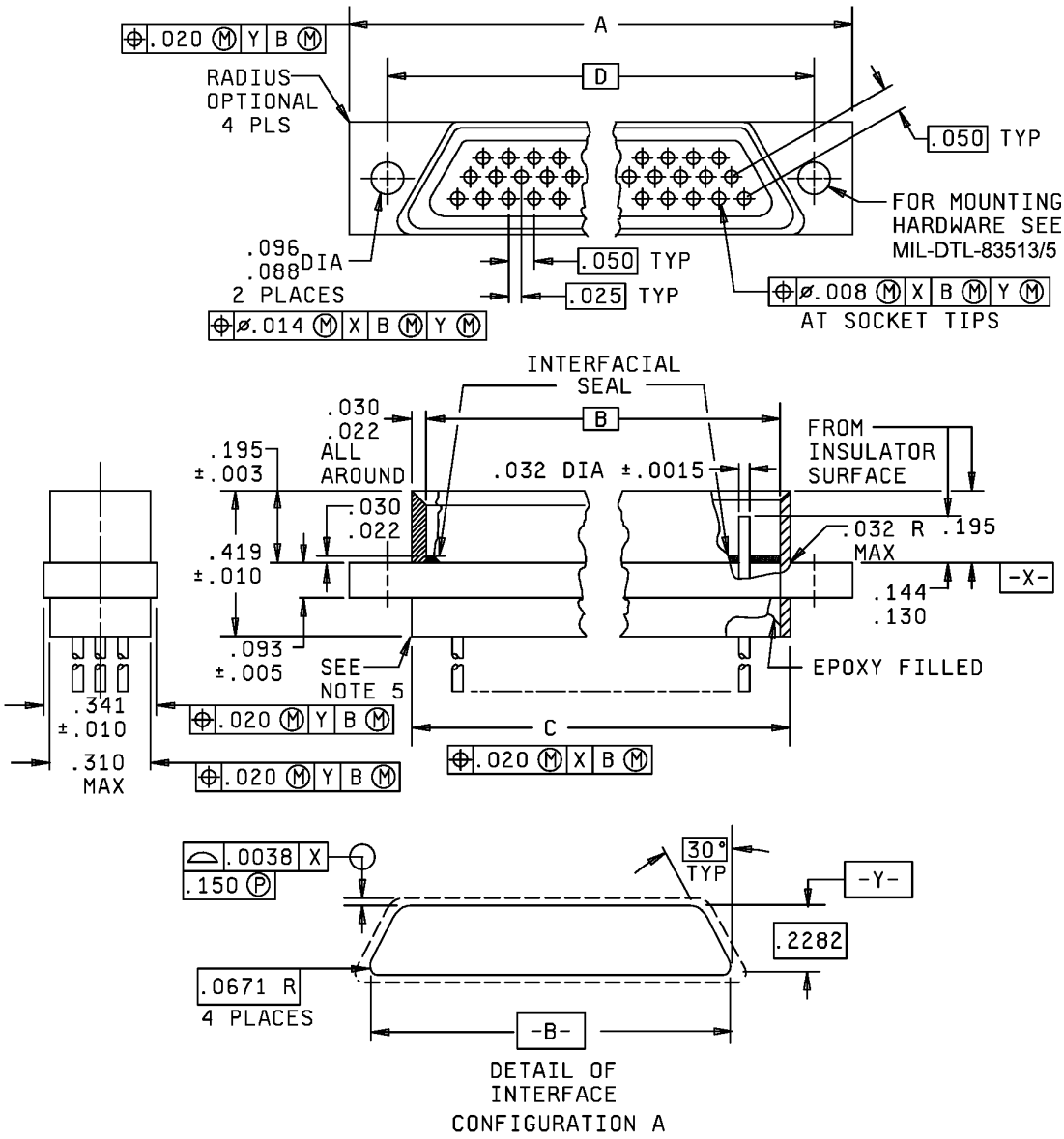


FIGURE 1. Connector, receptacle, .050 spacing - Continued.

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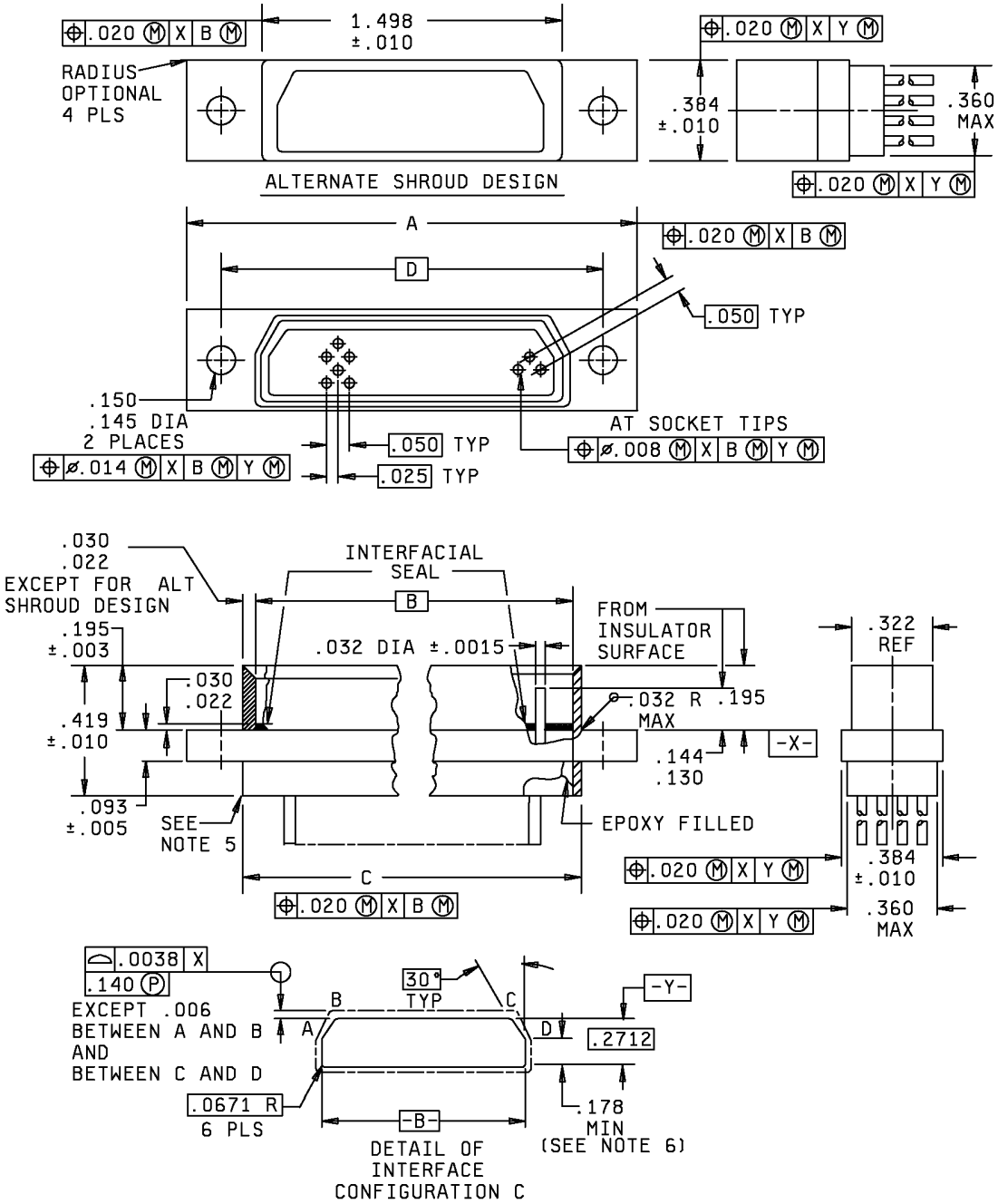


FIGURE 1. Connector, receptacle, .050 spacing - Continued.

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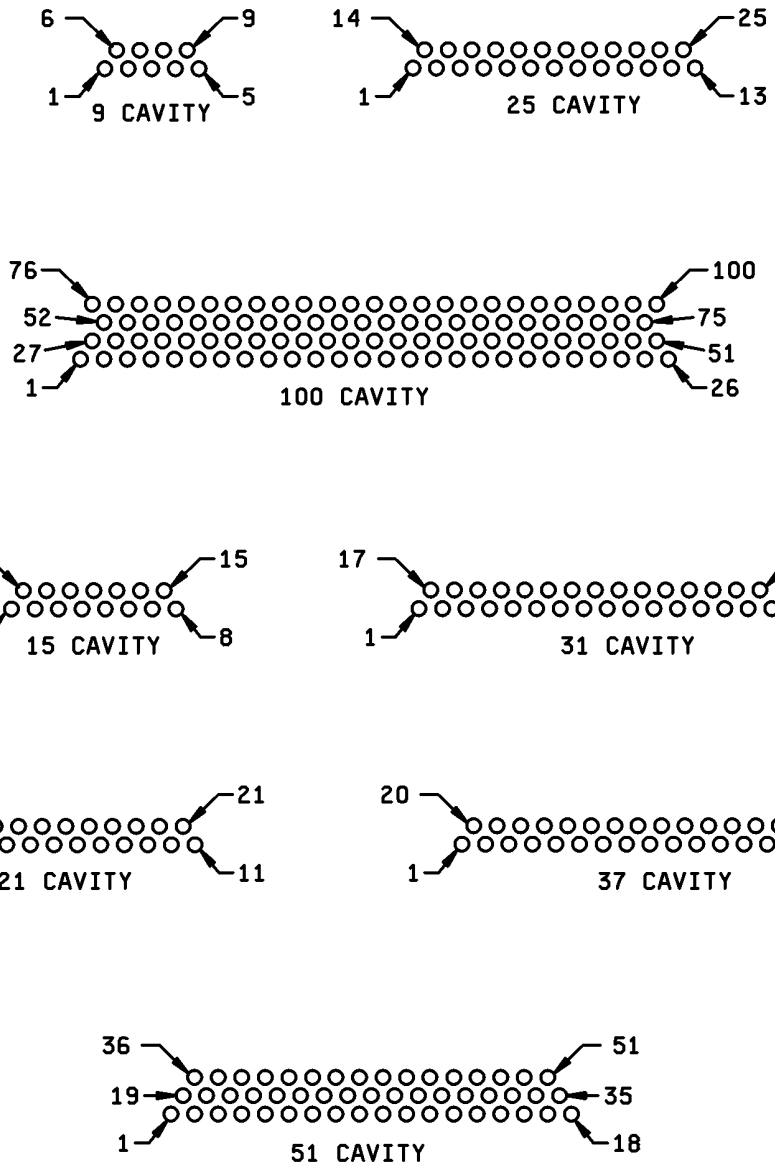
Number of cavities	A .010 (.25)	B BSC	C +.010 -.018 +(.25) -(.46)	D BSC	Configuration letter
9	.775 (19.68)	.3342 (8.489)	.390 (9.91)	.565 (14.35)	A
15	.925 (23.50)	.4842 (12.299)	.540 (13.72)	.715 (18.16)	A
21	1.075 (27.30)	.6342 (16.109)	.690 (17.53)	.865 (21.97)	A
25	1.175 (29.84)	.7342 (18.649)	.790 (20.07)	.965 (24.51)	A
31	1.325 (33.66)	.8842 (22.459)	.940 (23.88)	1.115 (28.32)	A
37	1.475 (37.46)	1.0342 (26.269)	1.090 (27.69)	1.265 (32.13)	A
51	1.425 (36.20)	.9842 (25.00)	1.040 (26.42)	1.215 (30.86)	B
100	2.160 (54.86)	1.3842 (35.159)	1.432 (36.37)	1.800 (45.72)	C

Inches	mm	Inches	mm
.002	0.05	.096	2.44
.003	0.08	.100	2.54
.0035	0.089	.1141	2.898
.0038	0.097	.117	2.97
.004	0.10	.123	3.12
.005	0.13	.130	3.30
.006	0.15	.144	3.66
.007	0.18	.145	3.68
.008	0.20	.150	3.81
.010	0.25	.180	4.57
.014	0.36	.1852	4.704
.018	0.46	.189	4.80
.020	0.51	.193	4.90
.022	0.56	.195	4.95
.023	0.58	.200	5.08
.025	0.64	.2282	5.796
.030	0.76	.232	5.89
.032	0.81	.239	6.07
.033	0.84	.270	6.86
.045	1.14	.2712	6.888
.050	1.27	.275	6.99
.067	1.702	.281	7.14
.0671	1.704	.298	7.57
.069	1.75	.310	7.87
.070	1.78	.322	8.18
.088	2.24	.341	8.66
.090	2.29	.360	9.14
.093	2.36	.384	9.75
.094	2.39	.419	10.64
.095	2.41	1.438	36.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. Metric equivalents are in parentheses or columns.
5 Point from which the lead length is measured.
6. Dimension line at point D locates the point of intersection of the two adjacent side of the receptacle interface.
7. 26 AWG wire is the maximum wire size that can be used in the solder cup.

FIGURE 1. Connector, receptacle, .050 spacing - Continued.

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NOTES:
1. Engaging face of pin insert shown.
2. Cavity identification numbers are for reference only and do not appear on part.

FIGURE 2. Insert arrangement.

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REQUIREMENTS:

Dimensions and configurations: See figure 1. This specification sheet describes the pin side of a rectangular metal shell connector. This connector uses reverse gender contact, i.e., the live pin recessed in the insulator with the static socket protruding from a shrouded interface.

Material and finish:

Shell finish: The requirements for shell finish shall be in accordance with MIL-DTL-83513.

Interfacial seal: Silicone rubber or fluorosilicone rubber or blend thereof.

Contact identification: See figure 2 for contact layout identification.

Contact connection: The pin contact (being recessed in the insulator) is normally connected to the live side of the circuit.

Current rating, maximum: 3 amperes per contact.

Mounting and mating hardware: Mounting hardware is to be ordered separately. For insert arrangements A through G, use MIL-DTL-83513/5, configurations A and B. For insert arrangement H, use MIL-DTL-83513/5, configuration C.

Mating plug: Shall conform to MIL-DTL-83513/1 or MIL-DTL-83513/3.

Part or Identifying Number (PIN): Consists of the letter M, the basic number of the specification sheet, a letter from the insert column and the shell finish.

M83513/04 - A 01 1/ C

M83513/04 - Specification sheet number

A - Insert arrangement (see figure 2)

A = 9
B = 15
C = 21
D = 25
E = 31
F = 37
G = 51
H = 100

01 1/ - Wire type

01 = M22759/11-26-9, 18 inches long 2/ 3/
02 = M22759/11-26-9, 36 inches long 2/ 3/
03 = M22759/11-26-( ), 18 inches long 3/ 4/ 5/
04 = M22759/11-26-( ), 36 inches long 3/ 4/ 5/
05 = A-A-59551, type S, 25 AWG, except shall be gold plated 10 microinches; 0.5 inch long 6/ 7/
06 = A-A-59551, type S, 25 AWG, except shall be gold plated 10 microinches; 1.0 inch long 6/ 7/
07 = A-A-59551, type S, 25 AWG, tin plated 100 microinches; 0.5 inch long 6/ 7/
08 = A-A-59551, type S, 25 AWG, tin plated 100 microinches; 1.0 inch long 6/ 7/
09 = M22759/33-26-9, 18 inches long 3/
10 = M22759/33-26-9, 36 inches long 3/
11 = M22759/33-26-( ), 18 inches long 3/ 4/
12 = M22759/33-26-( ), 36 inches long 3/ 4/
13 = M22759/11-26-9, 72 inches long 2/ 3/
14 = M22759/11-26-( ), 72 inches long 3/ 4/ 5/
15 = M22759/33-26-9, 72 inches long 3/
16 = M22759/33-26-( ), 72 inches long 3/ 4/

C - Shell finish (Interface critical)

C = Cadmium
N = electroless nickel (space applications only)
P = Passivated stainless steel

See footnotes on next page.

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1/ These connectors have leads attached of which wire type, wire size, wire color, and lead length are specified in PIN.
2/ For connectors with shell finish N intended for space applications, wire shall be M22759/33-26-9.
3/ Lead length tolerance +1.0, -0.0 inch.
4/ Color coding in accordance with MIL-STD-681, system I, except the appropriate color code shall be the individual contact cavity number minus 1 for contact positions 1 through 10 only, 11 through 20, etc. For example:

Pin number	Identification
4	3
9	8
21	0
31	0
77	6
100	9

5/ For connectors with shell finish N intended for space applications, wire shall be M22759/33-26-( ).
6/ Lead length tolerance +0.2, -0.0 inch.
7/ Contact and wire may be a single piece design.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AT, MI
Navy - AS, CG, MC, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project: 5935-4310-04)

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